Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Sommario

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Capitolo 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

POLITECNICO DI TORINO

Capitolo 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[a4paper,11pt,twoside,english]toptesi [latin1]inputenc [T1]fontenc lmodern [a-1b]pdfx hyperref cite listings color url multirow amsmath verbatim

Master Degree course in Communications and Computer Networks Engineering

Master Degree Thesis

The title of my MSc thesis.

Supervisors
Prof. Name1 Surname1
Prof. Name2 Surname2

Candidate
Your Name Your Surname

Academic Year 202*-202*

Acknowledgements

Write here you acknowledgments.

Abstract

Write here your abstract.

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Chapter 3

Introduction

In this chapter, you must provide an overview of your thesis, written such that a non-engineer person (one of your parents) should be able to read and understand at high level.

- what is the addressed scenario and the addressed problem?
- why the problem you addressed is interesting and/or practical relevant?
- what is you main contribution?
- which methodology you used? (theory, simulation, experiments, literature overview)?
- how the thesis is organized? which topic is covered in each chapter?

In the following the titles of chapters and sections are just examples.

3.1 Hints for writing

- Use short sentences.
- Never cut & paste a sentence from any website or paper. It is a copyright infringement and Politecnico anti-plagiarism system will detect it.

3.2 Latex hints

3.2.1 Latex examples

According to what has been discussed in Sec. 4.2, the following relation holds:

$$E = m2c^2 \tag{3.1}$$

Consider that eq:1 is contradicting [**?**].

With the above example, you have learned how to include math expressions, how to refer to equations, to other sections and to cite a related work.

3.2.2 Latex hints

- never use `\\` to go to a new line. Instead leave al least one line empty for a new line. Leave latex to use the proper formatting for the paragraph.
- use `~` for all cases you need a space that cannot be split across multiple lines. I.e., "`Sec.~\ref{xxx}`" and not "`Sec.~\ref{xxx}`", or "`according to~\eqref{xxx}`" and not "`according to \eqref{xxx}`".

3.3 Plagiarism

To be approved, your thesis must pass a plagiarism check. As a reminder:

- it is forbidden to copy verbatim one or more sentences from any other external source (web pages, papers, etc.), unless you explicitly grant the source and put the sentence/sentences in "...". Your can instead rephrase the whole sentence/sentences.
- it is forbidden to include an image (diagram or figure) taken from web pages or papers. It is better to redraw. If you really need to include an image from an external source, you must add to the caption "(Reproduced from [**?**])", citing properly the source.

Chapter 4

System model

This chapter should provide all the required information for an engineer who does not know anything about your thesis topic and would like to get the required background to understand your work and contribution

4.1 Model description

...

4.2 Related work

After having described the system model, you should discuss about the related papers. Each work should be described specifying the scenario, the provided main contributions and you should highlight the similarities and differences with respect to your work in this thesis.

Chapter 5

My main contribution

This chapter should be devoted to describe your main contribution, e.g., the algorithms you devised/studied.

Chapter 6

Experimental/numerical evaluation

Given the contribution in Chapter 5, you must provide the adopted methodology and the results

6.1 Methodology

Explain in all the details the adopted methodology, in order to make the results reproducible by anyone interested in addressing the same problem.

6.2 Numerical results

Here you will include all the main graphs. Each graph should appear with some explanation. The explanation must explain why the graph is interesting and what we learn from it, and not describing the numerical values you have on it (the reader is not blind and see by himself the numbers). Consider the example of Fig. 6.1.

Some basic hints:

- both x and y axes should be properly labeled. Units of measurements must be *always* reported!
- if the curve are experimental or by simulation, remember to report always all the points in addition to the line connecting them

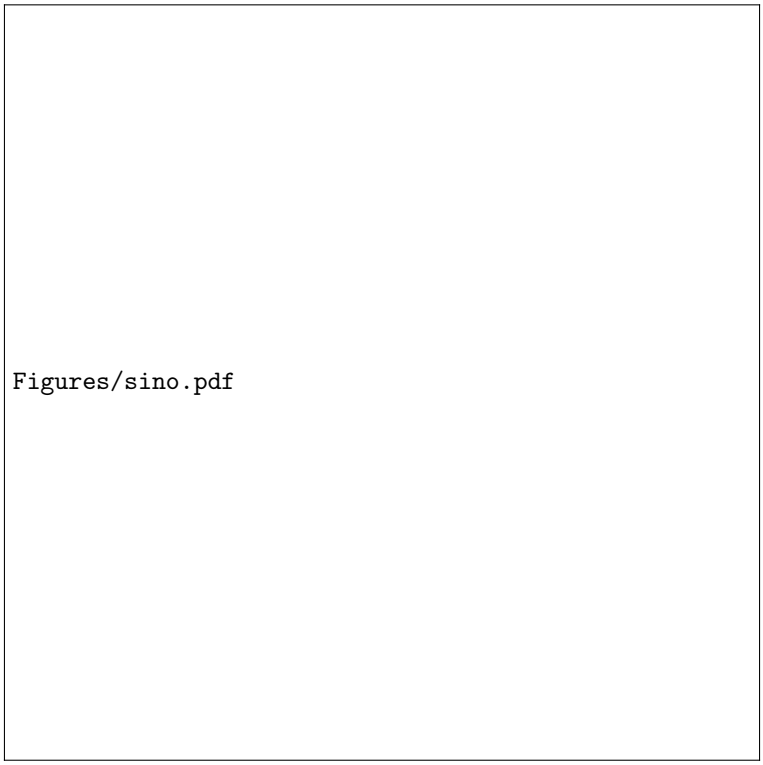


Figure 6.1. This is the required caption of the graph.

Chapter 7

Conclusion

Here you must summarize completely your work. You do not need anymore to motivate your topic, but just report in short what you have done and the main results.

If you wish, you can also add the possible direction for future works, extending your thesis work.